

Problem.

Prove that e is irrational.

Proof.

Suppose that $e = \frac{a}{b}$ for some $a, b > 0 \in \mathbb{N}$.

We know that the McLaurin's expansion for e^x is the following:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

So, for $e^1 = e$, we have that:

$$e^1 = \sum_{n=0}^{\infty} \frac{1}{n!} = e \quad (1)$$

Let us define x as the quantity:

$$x = b! \left(e - \sum_{n=0}^b \frac{1}{n!} \right) \quad (2)$$

Substitute $e = \frac{a}{b}$:

$$x = b! \left(\frac{a}{b} - \sum_{n=0}^b \frac{1}{n!} \right) = a(b-1)! - \sum_{n=0}^b \frac{b!}{n!} \quad (3)$$

Considering this new expression for x , it is trivial to see that x is an integer. That is because $a(b-1)!$ is an integer, and $\sum_{n=0}^b \frac{b!}{n!}$ is also an integer because $b \geq N$ for all $N \in [0, b] \cap \mathbb{N}$.

Any difference of integers is an integer by the definition of $\mathbb{Z}$:

$$\mathbb{Z} = \{a - b \mid a, b \in \mathbb{N}\}$$

Therefore $x \in \mathbb{Z}$.

However, from (1) and (2) it also follows that:

$$x = b! \left(\sum_{n=0}^{\infty} \frac{1}{n!} - \sum_{n=0}^b \frac{1}{n!} \right) = \sum_{n=b+1}^{\infty} \frac{b!}{n!}$$

Since the series is increasing and each term is positive, it follows that $x > 0$.

Let us notice that:

$$\frac{b!}{n!} = \frac{b \cdot (b-1) \cdot (b-2) \dots}{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (b+1) \cdot b \cdot (b-1) \cdot \dots} = \frac{1}{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (b+1)}$$

After the simplifications, the denominator has $n - b$ factors.

Since $n \geq b + 1$ for all $n \in [b + 1, \infty)$, the sum is bounded by:

$$\sum_{n=b+1}^{\infty} \frac{b!}{n!} < \sum_{n=b+1}^{\infty} \frac{1}{(b+1)^{n-b}} = \sum_{n=1}^{\infty} \left(\frac{1}{b+1} \right)^n = \frac{\frac{1}{b+1}}{1 - \frac{1}{b+1}} = \frac{1}{b} \leq 1 \quad (4)$$

since $b \in \mathbb{N}$.

This leads to conclude that

$$0 < x < 1$$

It has been established in (3) that x is an integer and it is possible to deduce from (4) that $x \in (0, 1)$. However, there are no integers in that interval.

A contradiction has been reached.

Q.E.D. ■